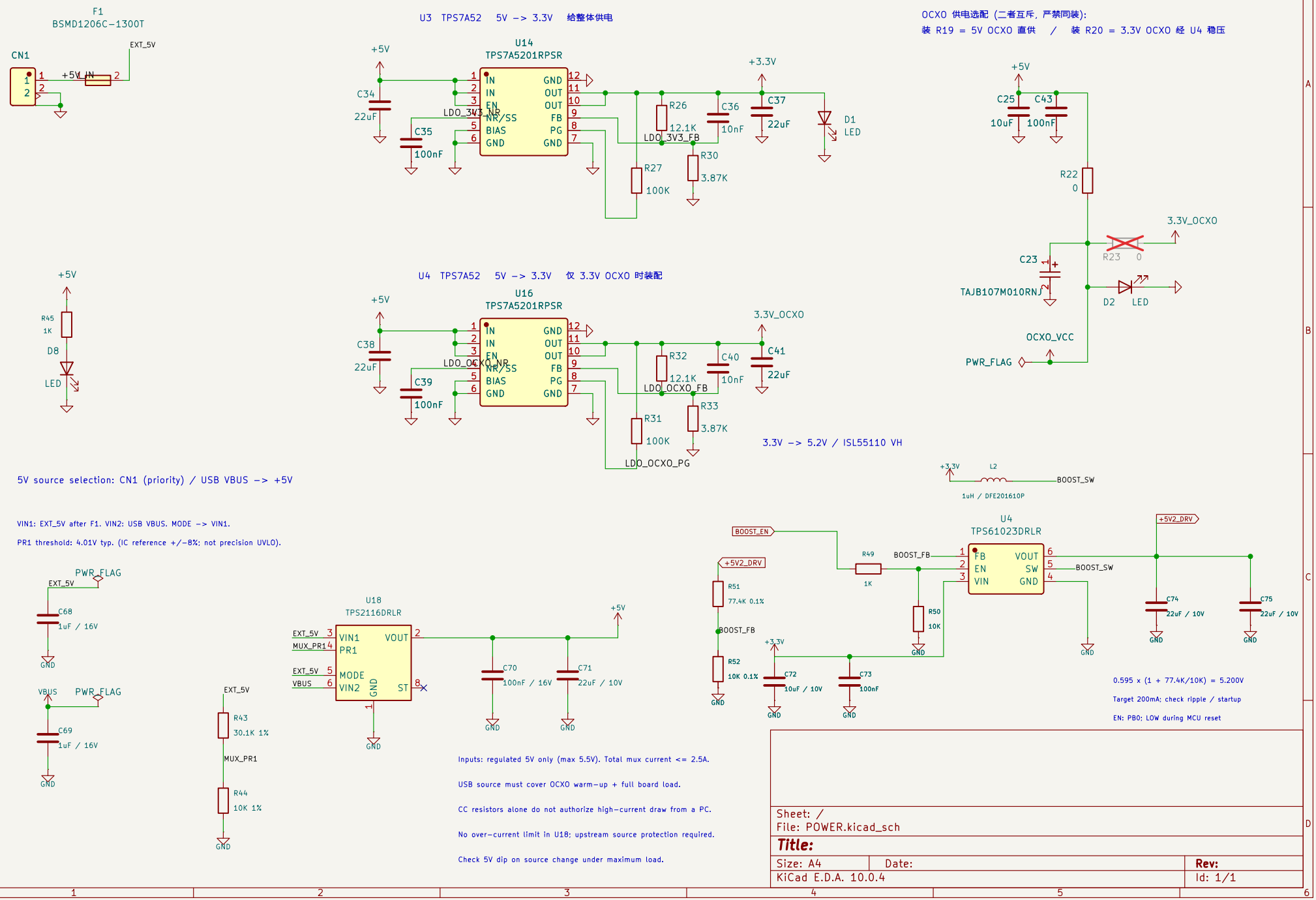

F1
BSMD1206C-1300T
CN1
EXT_5V
+5V_IN
U3 TPS7A52 5V -> 3.3V 给整体供电
U14
TPS7A5201RPSR
+5V
C34
22uF
C35
100nF
LDO_3V3_NR
IN
IN
EN
NR/SS
BIAS
GND
OUT
OUT
FB
PG
GND
R26
12.1K
LDO_3V3_FB
C36
10nF
R27
100K
R30
3.87K
C37
22uF
+3.3V
D1
LED
OCXO 供电选配（二者互斥，严禁同装）:
装 R19 = 5V OCXO 直供 / 装 R20 = 3.3V OCXO 经 U4 稳压
C25
10uF
C43
100nF
R22
0
3.3V_OCXO
R23
0
C23
TAJB107M010RNJ
D2
LED
OCXO_VCC
PWR_FLAG
U4 TPS7A52 5V -> 3.3V 仅 3.3V OCXO 时装配
U16
TPS7A5201RPSR
C38
22uF
C39
100nF
LDO_OCXO_NR
LDO_OCXO_FB
LDO_OCXO_PG
R32
12.1K
C40
10nF
R31
100K
R33
3.87K
C41
22uF
3.3V -> 5.2V / ISL55110 VH
R45
1K
D8
LED
5V source selection: CN1 (priority) / USB VBUS -> +5V
VIN1: EXT_5V after F1. VIN2: USB VBUS. MODE -> VIN1.
PR1 threshold: 4.01V typ. (IC reference +/-8%; not precision UVLO).
PWR_FLAG
C68
1uF / 16V
VBUS
C69
1uF / 16V
U18
TPS2116DRLR
VIN1
PR1
MODE
VIN2
VOUT
ST
MUX_PR1
R43
30.1K 1%
R44
10K 1%
C70
100nF / 16V
C71
22uF / 10V
L2
1uH / DFE201610P
BOOST_SW
U4
TPS61023DRLR
FB
EN
VIN
VOUT
SW
GND
BOOST_EN
+5V2_DRV
R49
1K
BOOST_FB
R51
77.4K 0.1%
R52
10K 0.1%
R50
10K
C72
10uF / 10V
C73
100nF
C74
22uF / 10V
C75
22uF / 10V
0.595 x (1 + 77.4K/10K) = 5.200V
Target 200mA; check ripple / startup
EN: PB0; LOW during MCU reset
Inputs: regulated 5V only (max 5.5V). Total mux current <= 2.5A.
USB source must cover OCXO warm-up + full board load.
CC resistors alone do not authorize high-current draw from a PC.
No over-current limit in U18; upstream source protection required.
Check 5V dip on source change under maximum load.
Sheet: /
File: POWER.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1